

①

$$\vec{u} \cdot \vec{v} = -1 \cdot 4 + 2 \cdot 6 = 8$$

$$\vec{u} \cdot \vec{u} = 1 + 4 = 5$$

$$\frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} = \frac{8}{5} = 1,6$$

② $\frac{1}{\vec{w} \cdot \vec{w}} \cdot \vec{w}$

$$\vec{w} \cdot \vec{w} = 8 + 1 + 25 = 35$$

$$\vec{w} = \begin{bmatrix} \frac{3}{35} \\ -\frac{1}{35} \\ -\frac{5}{35} \end{bmatrix}$$

③

$$\left(\frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \right) \vec{v} = \begin{bmatrix} \frac{2}{13} \cdot 4 \\ \frac{2}{13} \cdot 6 \end{bmatrix} = \begin{bmatrix} \frac{8}{13} \\ \frac{12}{13} \end{bmatrix}$$

$$\vec{v} \cdot \vec{v} = 16 + 36 = 52$$

$$\frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} = \frac{8}{52} = \frac{2}{13}$$

$$|w| = \sqrt{8+1+25} = \sqrt{34}$$

$$\begin{bmatrix} 30 \\ 40 \end{bmatrix}$$

$$u = \begin{bmatrix} -30 \\ 40 \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

$$|u| = \sqrt{9+16} = 5$$

$$\hat{u} = \begin{bmatrix} -3/5 \\ 4/5 \end{bmatrix}$$

(12)

$$\hat{u} = \begin{bmatrix} 7/9 \\ 1/9 \\ 1 \end{bmatrix}$$

$$|u| = \sqrt{\frac{49}{81} + \frac{1}{81} + 1} = \sqrt{\frac{89}{81}} = \frac{\sqrt{89}}{9}$$

$$\hat{u} = \begin{bmatrix} 7/\sqrt{89} \\ 1/\sqrt{89} \\ 9/\sqrt{89} \end{bmatrix}$$

(13)

$$x = \begin{bmatrix} 10 \\ -3 \end{bmatrix}$$

$$y = \begin{bmatrix} -1 \\ -5 \end{bmatrix}$$

$$d = \sqrt{(10-(-1))^2 + (-3-(-5))^2} = \sqrt{11^2 + 2^2} = \sqrt{125} = 5\sqrt{5}$$

(14)

$$\bar{a} = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$$

$$\bar{b} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$$

$$\bar{a} \cdot \bar{b} = 8 \cdot (-2) + (-5) \cdot (-3) = -1$$

not orthogonal

(17)

$$\bar{u} \cdot \bar{v} = 3 \cdot (-4) + 2 \cdot 1 + (-5) \cdot (-2) + 0 \cdot 6 = 0$$

$\bar{u}$ and $\bar{v}$ are orthogonal $\bar{u} \perp \bar{v}$

(18)

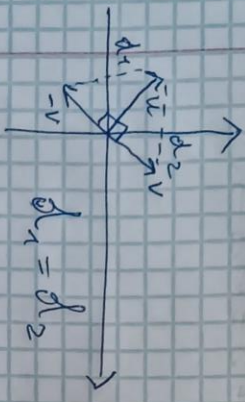
$$a) \bar{v} \cdot \bar{v} = |\bar{v}|^2 \quad \text{True} \quad \text{let } v = \begin{bmatrix} a \\ b \end{bmatrix} \quad v \cdot v = a^2 + b^2 \text{ and } |\bar{v}| = \sqrt{a^2 + b^2}$$

$$|v| = \sqrt{a^2 + b^2} \quad \text{which is } \sqrt{v \cdot v} \text{ so } v \cdot v = |v|^2$$

b) $\bar{u} \cdot (c\bar{v}) = c(\bar{u} \cdot \bar{v})$ True, scalars don't change the direction, they just change the length

3

c) $\text{dist}(u, v) = \text{dist}(u, -v)$
 True



d)

for $A \in \mathbb{R}^2$, $\text{col } A \perp \text{Nul } A$
 False
 $(\text{Col } A)^\perp = \text{Nul } (A^T)$

e) if $v_1, \dots, v_n$ span a subspace W ,
 if $x \perp \text{each } v_i$, then x is in $W^\perp$
 True

21)

$$(u+v)w = (u+v)^T w = (u^T + v^T)w = u^T w + v^T w = w^T u + w^T v = w^T(u+v) = (u+v)^T w$$

23)

$$u = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix} \quad v = \begin{bmatrix} -4 \\ -4 \\ 6 \end{bmatrix}$$

$$u \cdot v = 2(-4) + 20 - 6 = 0$$

$$u \cdot u = 2^2 + (-3)^2 + (-1)^2 = 14 \neq 0$$

$$v \cdot v = (-4)^2 + (-4)^2 + 6^2 = 68 \neq 0$$

$$u \cdot v = 2(-4) + 20 - 6 = 0$$

$$|u+v|^2 = (u+v) \cdot (u+v) = 14 + 68 + 2(-4)(-4) + 2(20)(-6) + 2(-1)(6) = 131$$

25)

$$v = \begin{bmatrix} a \\ 8 \end{bmatrix}$$

$$u = \begin{bmatrix} x \\ y \end{bmatrix}$$

orthogonal to v

$$\text{if } a=0, b=0 \Rightarrow 0x+0y=0 \quad u \in \mathbb{R}^2$$

$$\text{if } a=0, b \neq 0 \Rightarrow by=0, y=0 \quad u = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

$$\text{if } a \neq 0 \Rightarrow x = -\frac{b}{a}y, u = \begin{bmatrix} -b \\ a \end{bmatrix}$$